METU
NORTHERN CYPRUS CAMPUS

Middle East Technical University

Northern Cyprus Campus

EEE 202 Design Project Presentation

Project 6: Water Level Indicator

Conductive-Probe-Based Discrete Analog Water Level Indicator Circuit

Tunay Arıcı	**Mehmet Yamaç Güngör**	**Arda Ardıç**	**Yiğitalp Dellal**
2688141	2585719	2689933	2585644

Outline

Motivation, Applications, and Design Requirements

Implemented Circuit Architecture and Operation

Component Selection and Design Calculations

LTspice Verification, Practical Results, and Improvement Directions

Motivation, Applications, and Design Requirements

Project Statement and Design Requirements

Given design problem

Design a discrete analog water level indicator circuit that detects the water level inside a container and displays the reached level using LED indicators.

Water height $\rightarrow$ electrical detection $\rightarrow$ visible LED output

Design parameter	**Requirement**
Supply voltage	$5\,\mathrm{V} \leq V_S \leq 12\,\mathrm{V}$
Detected levels	At least 3 water levels
Output indicator	LEDs
Sensing method	Conductive probes
Circuit type	Discrete analog implementation

Main requirement interpretation

Motivation, Applications, and Historical Context

Why level detection matters

- Preventing tank overflow
- Avoiding dry-running of pumps
- Monitoring stored liquid level
- Providing alarm indication

Typical application areas

- Domestic water tanks
- Reservoirs and storage tanks
- Wastewater systems
- Automatic filling systems

Visual inspection $\rightarrow$ mechanical float $\rightarrow$ electrical probe $\rightarrow$ industrial transmitter

Connection to this project

Our design is an educational discrete analog version of electrical probe-based level detection.

Industrial Applications: Companies and Devices

Company	Example device type	Typical application
Endress+Hauser	Conductive, radar, ultrasonic, hydrostatic sensors	Tanks, silos, wastewater
VEGA	VEGAPULS radar, VEGAFLEX guided radar sensors	Process tanks, storage tar
Emerson Rosemount	Radar and guided-wave radar transmitters	Industrial level measureme

Industrial meaning

Industrial level measurement systems use different sensing principles depending on the liquid, tank geometry, safety requirements, and operating environment.

Position of our project

Our circuit is not an industrial transmitter. It is a low-cost educational prototype that models the same basic task: detecting liquid level and producing an output.

Implemented Design Choices

Minimum project requirement

- $5\,\mathrm{V}$ to $12\,\mathrm{V}$ supply
- Minimum 3 detected levels
- LED-based indication
- Conductive-probe sensing

Our implemented design

- $12\,\mathrm{V}$ DC supply
- Six sensing levels
- First five LEDs: steady indication
- Highest level: blinking alarm

Minimum 3 levels $\longrightarrow$ 6-level indicator with alarm behavior

Design decision

The highest water level was separated from normal level indication and used as a warning stage instead of only another steady LED.

Modular Design Flow and Design Evolution

Conductive probe → base drive → BJT switch → LED output

input stabilization → supply protection → astable alarm stage

Version	Main feature	Design role
V1	Five-level BJT indicator	Basic sensing principle
V2	Sixth level + blinking alarm	Critical-level warning
V3	Stabilization + protection	Final implemented design
V4	MOSFET-based sensing	Lower-current improvement
V5	AC-based sensing	Corrosion-reduction direction

Presentation focus

The main focus is Version 3: a BJT-based discrete analog circuit with stabilization, protection, and a blinking alarm stage.

Implemented Circuit Architecture and Operation

Implemented Version 3 Architecture

Main implemented design

Version 3 is the implemented laboratory circuit. It uses conductive probes, BJT switching stages, LED indicators, input stabilization, reverse-polarity protection, and an astable alarm stage.

Water level $\rightarrow$ probe contact $\rightarrow$ BJT switching $\rightarrow$ LED indication

Protection

- 1N5819
- Reverse-polarity protection
- Protected V_{CC}

Repeated sensing

- Ordinary probes P_n
- Water resistance
- BC547B stages

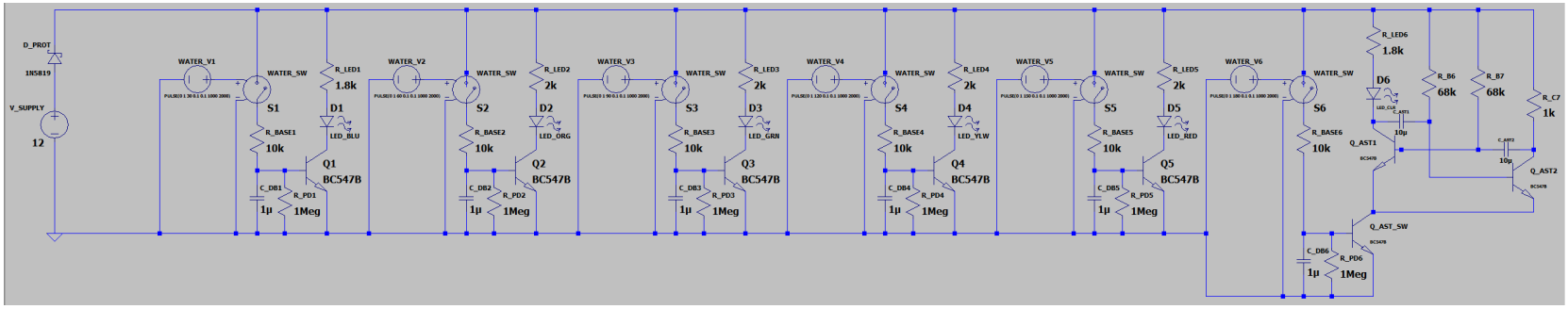
Alarm stage

- Alarm probe P_A
- Astable multivibrator
- Blinking LED D_A

Circuit structure

The ordinary levels use repeated modules indexed by n, while P_A

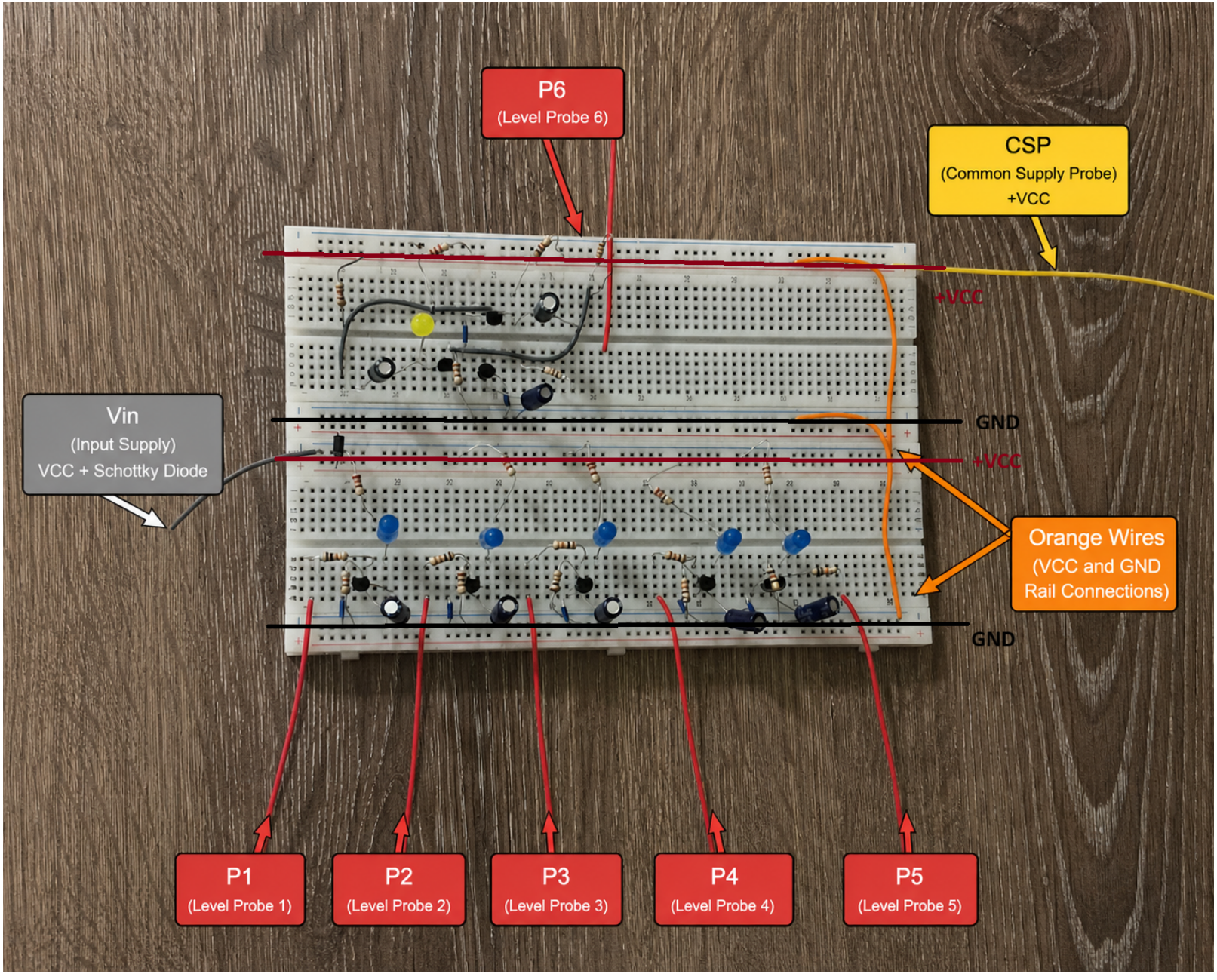
Implemented Version 3 Schematic



How to read this schematic

The left side contains the protected supply input, the middle contains the repeated ordinary sensing-and-indication stages, and the right side contains the alarm-level astable stage.

Physical Implementation of the Circuit



Conductive Probe Sensing and Water Resistance

Sensing principle

When water touches the common supply probe and a level probe P_n, it creates a conductive path that drives the corresponding BJT stage.

$$V_{CC} \rightarrow \text{common probe} \rightarrow R_{\text{water},n} \rightarrow P_n \rightarrow R_{\text{BASE},n} \rightarrow \text{BJT base}$$

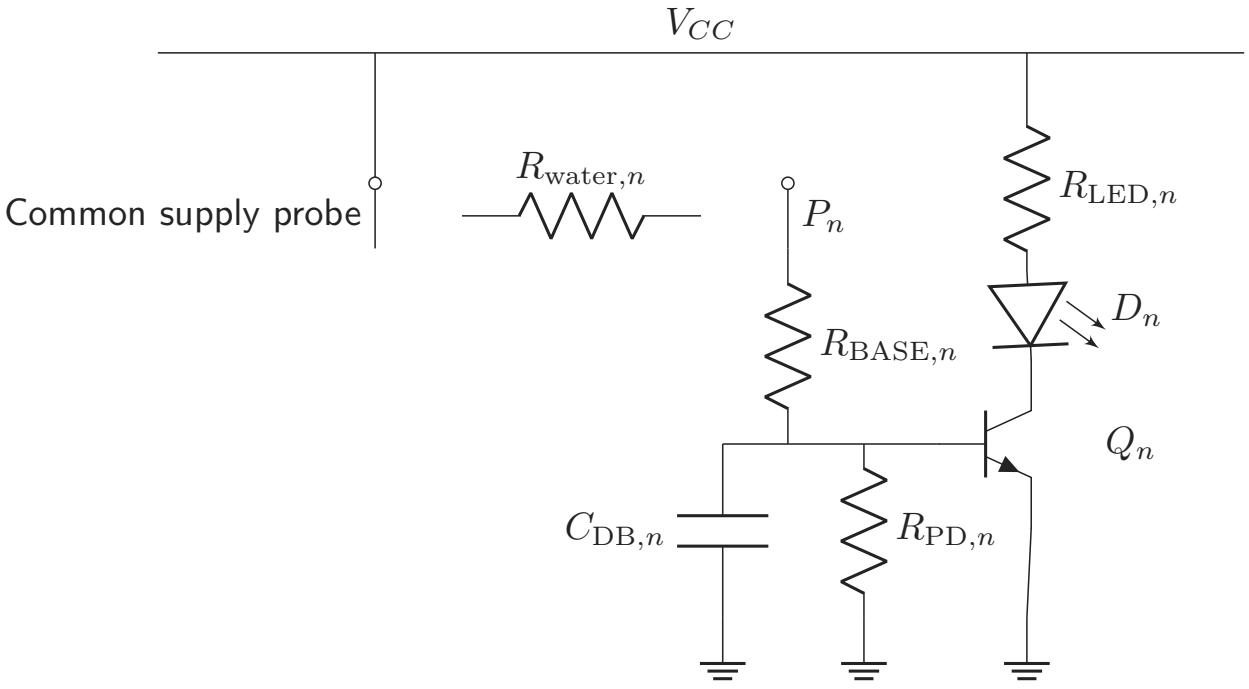
$$\text{Water reaches } P_n \rightarrow I_{B,n} \text{ flows} \rightarrow Q_n \text{ ON} \rightarrow D_n \text{ ON}$$

$$R_{\text{water},n} = \rho_{\text{water}} \frac{\ell}{A} = \frac{\ell}{\sigma_{\text{water}} A} \qquad R_{\text{water},n} \uparrow \Rightarrow I_{B,n} \downarrow \Rightarrow \text{weaker BJT drive}$$

Important physical point

The water path is not an ideal wire. It depends on water conductivity, ion concentration, temperature, probe spacing, and effective contact

Repeated Sensing and Indication Module



$$P_n \text{ reached} \rightarrow Q_n \text{ ON} \rightarrow D_n \text{ ON}, \qquad n = 1, \ldots, 5$$

Modular meaning

This is the repeated unit of the ordinary level indicators. More levels can be added by repeating the same probe–BJT–LED structure.

Alarm-Level Module and Astable Behavior

Why P_A is different

The ordinary probes P_n, where $n = 1, \ldots, 5$, provide normal level indication. The alarm-level probe P_A represents the highest water condition and activates the blinking alarm behavior.

$$P_A \text{ reached} \rightarrow Q_{\mathrm{AST_SW}} \text{ ON} \rightarrow \text{astable enabled} \rightarrow D_A \text{ blinks}$$

Ordinary levels

- $P_n,\ n = 1, \ldots, 5$
- Steady LED indication
- Cumulative level display

Alarm level

- P_A
- Highest water condition
- Blinking warning output

Design reason

The highest level is not only information; it is a warning condition. A blinking LED is more noticeable than another steady LED.

Input Stabilization and Protection

Pull-down resistor

- Prevents floating base nodes
- Keeps dry probe OFF

$V_{B,n} \approx 0 \Rightarrow Q_n$ OFF

Debounce capacitor

- Smooths fast contact changes
- Reduces flickering

water motion $\rightarrow$ smoothed input

Protection diode

- $1\mathrm{N}5819$
- Series supply protection
- Blocks reverse polarity

$V_{CC,\mathrm{eff}} \approx 11.7\,\mathrm{V}$

Practical role

These elements make the circuit more suitable for physical implementation, not only for ideal simulation.

Sequential Operation Summary

Water level condition	**Circuit response**
Below P_1	All LEDs OFF
Water reaches P_1	D_1 ON
Water reaches P_2	D_1, D_2 ON
Water reaches P_3	D_1, D_2, D_3 ON
Water reaches P_4	D_1 to D_4 ON
Water reaches P_5	D_1 to D_5 ON steadily
Water reaches P_A	D_A blinks as alarm

Main operational feature

The circuit separates ordinary cumulative level indication from alarm-level blinking behavior.

BJT Limitation and MOSFET Improvement Direction

Implemented BJT sensing

$$I_{B,n} \approx \frac{V_{CC} - V_{BE}}{R_{\text{water},n} + R_{\text{BASE},n}}$$

$$I_{B,n} \gtrsim \frac{I_{C,n}}{\beta_{\text{forced}}}$$

$$I_{\text{LED},n} \approx \frac{V_{CC} - V_{F,n} - V_{CE,n}}{R_{\text{LED},n}}$$

If $R_{\text{water},n}$ increases, base current decreases and the BJT may be driven more weakly.

MOSFET-based improvement

$$I_G \approx 0$$

$$V_{GS} > V_{\text{th}}$$

$$V_{DS(\text{on})} \approx I_D R_{DS(\text{on})}$$

The water path mainly establishes gate voltage instead of supplying continuous base current.

Design insight

The BJT version was implemented because it is simple and available. A properly protected MOSFET version can reduce sensing current, reduce corrosion risk, and make switching less dependent on water-path current.

Component Selection and Design Calculations

Design Strategy and Main Assumptions

Main purpose

The component values were selected to make the circuit visible, stable, low-power, and suitable for laboratory implementation.

Supply → LED current → BJT switching → stabilization → alarm timing → pow

Quantity	Assumed value
Input supply voltage	$V_S = 12\,\mathrm{V}$
Schottky diode drop	$V_D \approx 0.3\,\mathrm{V}$
Effective supply	$V_{CC,\mathrm{eff}} \approx 11.7\,\mathrm{V}$
BJT base-emitter voltage	$V_{BE} \approx 0.7\,\mathrm{V}$
BJT saturation voltage	$V_{CE(\mathrm{sat})} \approx 0.2\,\mathrm{V}$
Modeled water resistance	$R_{\mathrm{water},n} = 100\,\mathrm{k\Omega}$
Target LED current	$I_{\mathrm{LED}} \approx 5\,\mathrm{mA}$

Presentation focus

Only key equations and design values are shown here; detailed

Component Roles and Datasheet Checks

Selected components

Component	Role
1N5819	Reverse protection
BC547B	LED switching
LEDs D_n, D_A	Visual outputs
LED resistors	Current limiting
Base resistors	Base-current limiting
$1\,\mathrm{M\Omega}$	Pull-down network
$1\,\mu\mathrm{F}$	Input stabilization
$68\,\mathrm{k\Omega}, 10\,\mu\mathrm{F}$	Astable timing

Suitability checks

Part	Value	Check
BC547B	$V_{CEO} = 45\,\mathrm{V}$	$> 12\,\mathrm{V}$
BC547B	$I_C = 100\,\mathrm{mA}$	$> 5\,\mathrm{mA}$
1N5819	$V_{RRM} = 40\,\mathrm{V}$	$> 12\,\mathrm{V}$
1N5819	$I_F \approx 1\,\mathrm{A}$	$> I_{\mathrm{total}}$
Supply	0–25 V	supports 12 V

Engineering purpose

Component ratings were checked to ensure that the selected parts operate below their main voltage and current limits.

Supply and LED Current Design

Supply and protection

$$5\,\mathrm{V} \le V_S \le 12\,\mathrm{V}$$

$$V_S = 12\,\mathrm{V}$$

$$V_{CC,\mathrm{eff}} \approx 12\,\mathrm{V} - 0.3\,\mathrm{V} = 11.7\,\mathrm{V}$$

- 1N5819 Schottky diode
- Lower drop than silicon diode
- Reverse-polarity protection

LED resistor design

$$I_{\mathrm{LED}} \approx 5\,\mathrm{mA}$$

$$R_{\mathrm{LED},n} \approx \frac{V_{CC,\mathrm{eff}} - V_{F,n} - V_{CE(\mathrm{sat})}}{I_{\mathrm{LED},n}}$$

LED	$\mathbf{V_F}$	$\mathbf{R_{LED}}$	$\mathbf{I_{LED}}$
Blue	3.2 V	1.8 kΩ	4.61 mA
Orange	2.0 V	2.0 kΩ	4.75 mA
Green	2.1 V	2.0 kΩ	4.70 mA
Yellow	2.0 V	2.0 kΩ	4.75 mA
Red	1.9 V	2.0 kΩ	4.80 mA
Clear/D_A	3.2 V	1.8 kΩ	4.61 mA

Brightness note

Similar current does not guarantee identical perceived brightness, because LED color, efficiency, and manufacturer differences also matter.

BJT Switching and Saturation Check

Modeled switching condition

$$R_{\text{water},n} = 100\,\text{k}\Omega$$

$$I_{B,n} \approx 0.1\,\text{mA}$$

$$I_{C,n} \approx I_{\text{LED},n} \approx 5\,\text{mA}$$

$$\beta_{\text{required}} = \frac{I_{C,n}}{I_{B,n}} \approx \frac{5\,\text{mA}}{0.1\,\text{mA}} = 50$$

Forced current gain check

$$\beta_{\text{forced}} = \frac{I_C}{I_B}$$

For conservative switch design,

$$\beta_{\text{forced}} \approx 10$$

$$I_{B,\text{forced}} \approx \frac{5\,\text{mA}}{10} = 0.5\,\text{mA}$$

$$I_{\text{LED},n} \approx \frac{V_{CC,\text{eff}} - V_{F,n} - V_{CE,n}}{R_{\text{LED},n}}$$

Interpretation

The BJT can switch under the modeled water condition, but the available base current does not guarantee deep saturation for all real water samples.

Input Stabilization and Alarm Timing

Input stabilization

$$R_{\mathrm{PD},n} = 1\,\mathrm{M\Omega}$$

$$C_{\mathrm{DB},n} = 1\,\mu\mathrm{F}$$

$$\tau = R_{\mathrm{eq}}C$$

$$\tau_{\mathrm{on}} \approx 0.11\,\mathrm{s}$$

$$\tau_{\mathrm{off}} \approx R_{\mathrm{PD},n}C_{\mathrm{DB},n} = 1.0\,\mathrm{s}$$

Alarm timing for D_A

For a symmetric BJT astable,

$$f \approx \frac{1}{1.386RC}$$

$$R = 68\,\mathrm{k\Omega}, \qquad C = 10\,\mu\mathrm{F}$$

$$f \approx 1.06\,\mathrm{Hz}$$

$D_A \rightarrow$ approximately one blink per second

Engineering interpretation

The input network suppresses brief contact disturbances, while the astable timing produces a clearly visible alarm blink.

Current, Power, and Main Practical Limitation

Estimated current

$$I_{\text{total}} \approx 30\,\text{mA} - 40\,\text{mA}$$

- Several LEDs may be ON together
- Astable stage adds extra current
- Still below supply limits

Estimated power

$$P_{\text{total}} \approx V_{CC,\text{eff}}\, I_{\text{total}}$$

$$0.35\,\text{W} \lesssim P_{\text{total}} \lesssim 0.47\,\text{W}$$

$P_{\text{total}} < 0.5\,\text{W} \Rightarrow$ low thermal stress

Main practical limitation

The circuit is not mainly limited by heating. More important concerns are water conductivity variation, DC sensing current, electrolysis, and probe corrosion.

From Design Values to LTspice Verification

What was established

The main component values were selected from the supply range, target LED current, BJT switching condition, input stabilization behavior, alarm timing, and power budget.

selected component values $\rightarrow$ expected circuit behavior $\rightarrow$ LTspice verification

Next step

After selecting the values, the complete Version 3 circuit must be verified in LTspice: sequential LED operation, alarm triggering, blinking behavior, current, and power.

LTspice Verification, Practical Results, and Improvement Directions

LTspice Verification Approach

Purpose of simulation

LTspice was used to verify whether the complete Version 3 circuit produces the intended electrical behavior before interpreting the physical experiment.

modeled water contact → BJT switching → LED response → alarm behavior → supply-side be

Simulation model

- Water contact modeled by controlled switches
- Water path modeled resistively
- Probes activated sequentially

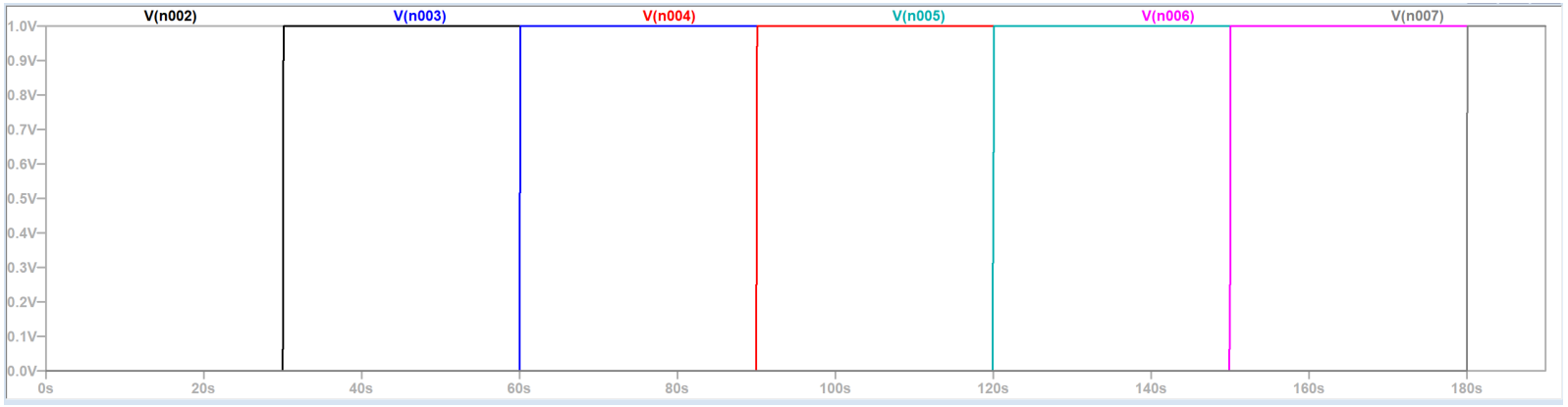
Observed quantities

- Trigger signals and base nodes
- BJT collector behavior
- LED and alarm currents
- Supply voltage, current, and power

Notation

In this presentation, P_A corresponds to the physical sixth probe P_6, and D_A corresponds to the alarm LED D_6.

Simulation 1: Sequential Water-Level Triggering



What the simulation shows

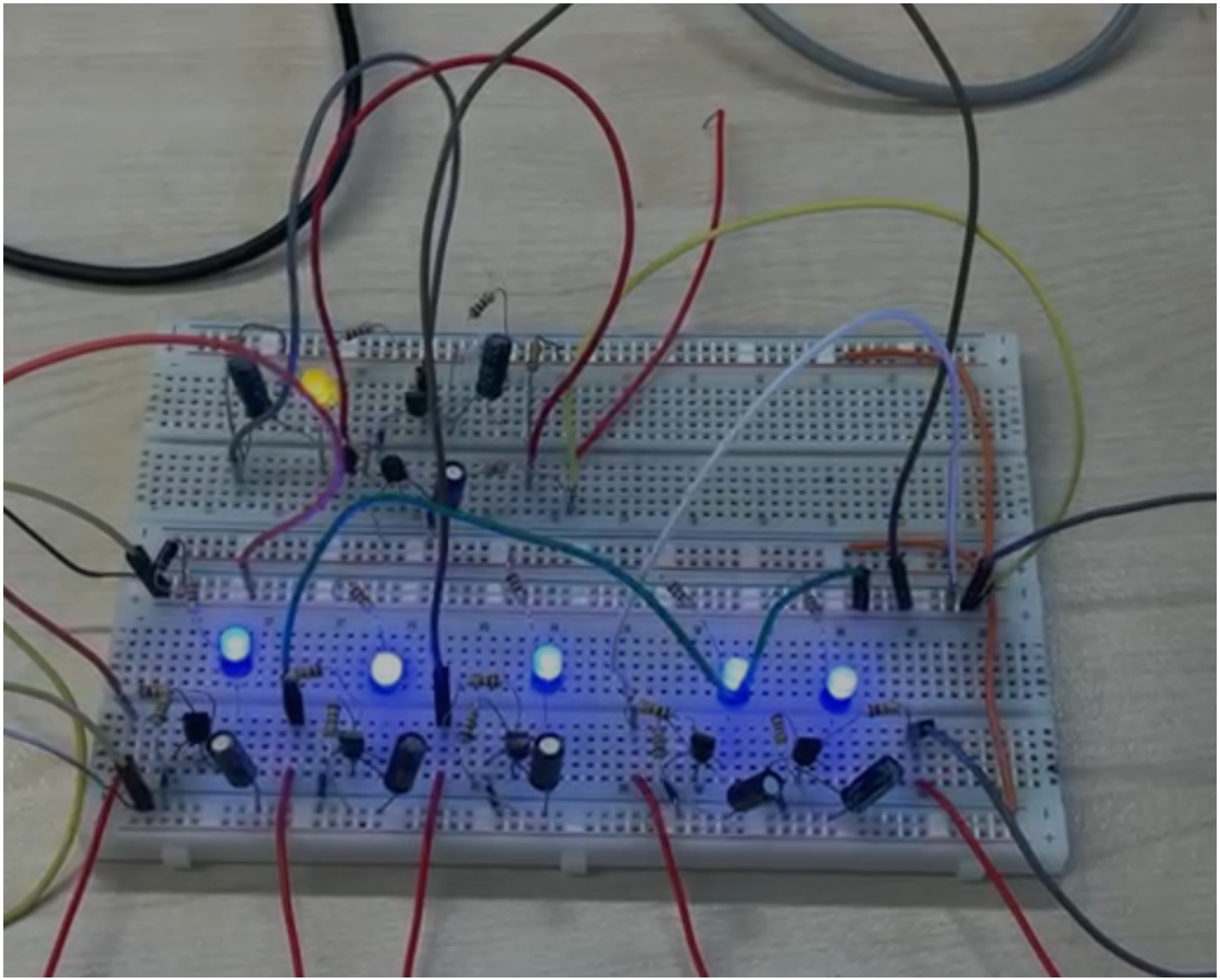
The switch-control signals activate one after another, representing a rising water level that reaches the probes sequentially.

$$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow P_A$$

Physical meaning

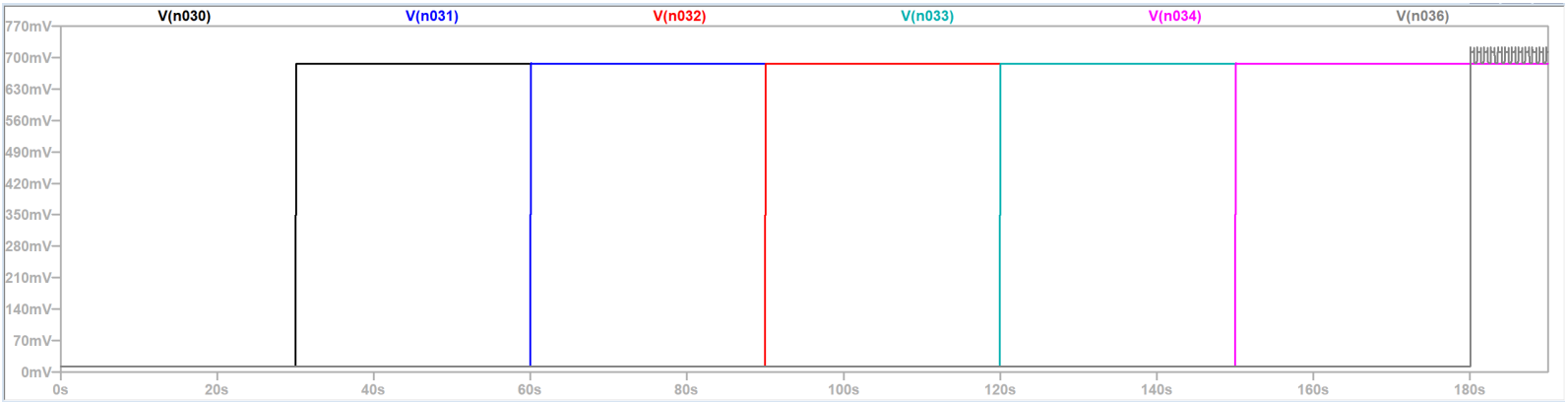
This is the simulated input sequence. It means the following plots can be interpreted as the circuit response to a gradually rising water level.

Physical Result: Sequential Level Indication



Experimental observation

Simulation 2: Input Triggering and RC Delay



What the simulation shows

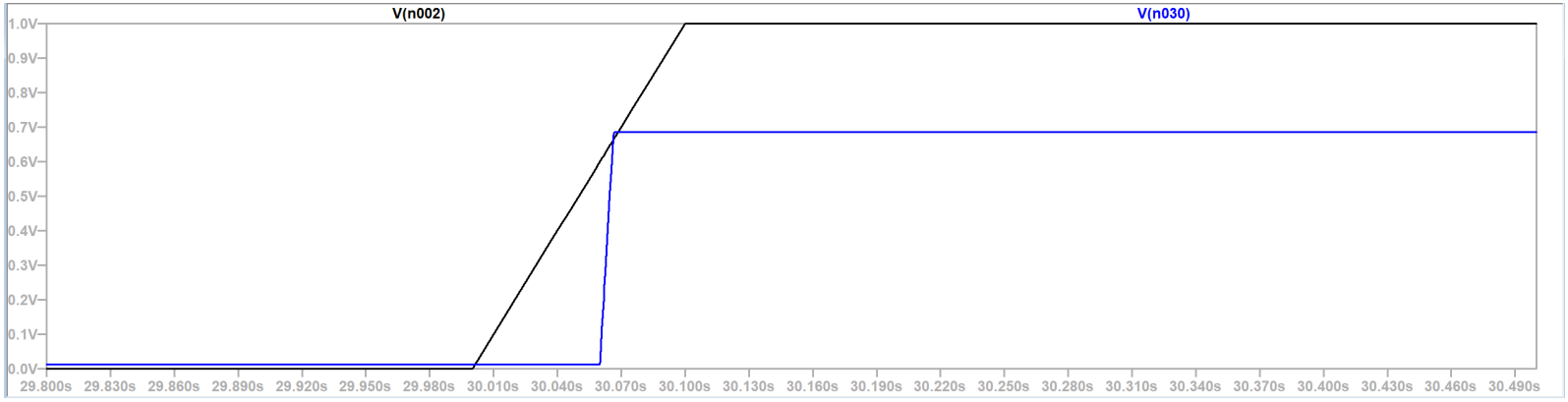
The BJT base-node voltages rise after their corresponding water-contact switches are activated. The first five base nodes settle near the expected silicon junction voltage.

$$V_{B,n} \approx 0.68\text{–}0.69\,\mathrm{V} \quad \Rightarrow \quad \text{base-emitter junction forward biased}$$

Physical meaning

Modeled water contact is successfully converted into BJT base drive. The RC network also slows sudden base-node changes, giving input

Simulation 3: BJT Switching Verification



What the simulation shows

When a sensing stage is activated, the corresponding BJT collector voltage drops close to ground.

Q_n ON $\Rightarrow$ $V_{CE,n}$ becomes small $\Rightarrow$ LED current path is completed

Physical meaning

The BJTs behave as low-side switches. They do not generate the LED voltage; they complete the path from LED branch to ground.

Physical Result: BJT Switching Measurement



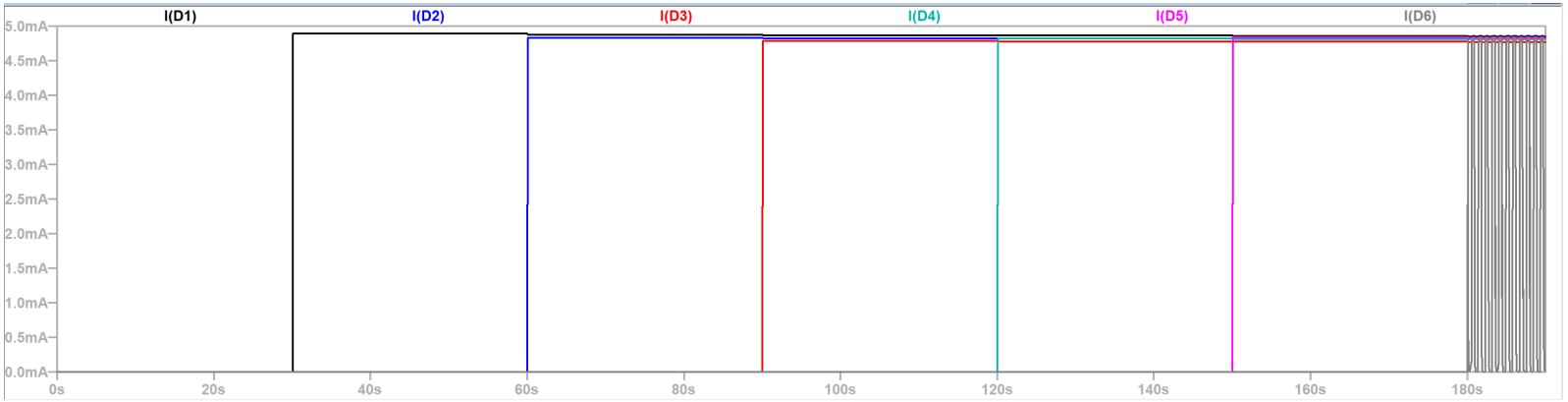
Measured switching behavior

The ON-state base-emitter voltage was measured around

$$V_{BE,\mathrm{ON}} \approx 0.729\,\mathrm{V},$$

and the ON-state collector-emitter voltage dropped to a very small

Simulation 4: LED Currents and Sequential Indication



What the simulation shows

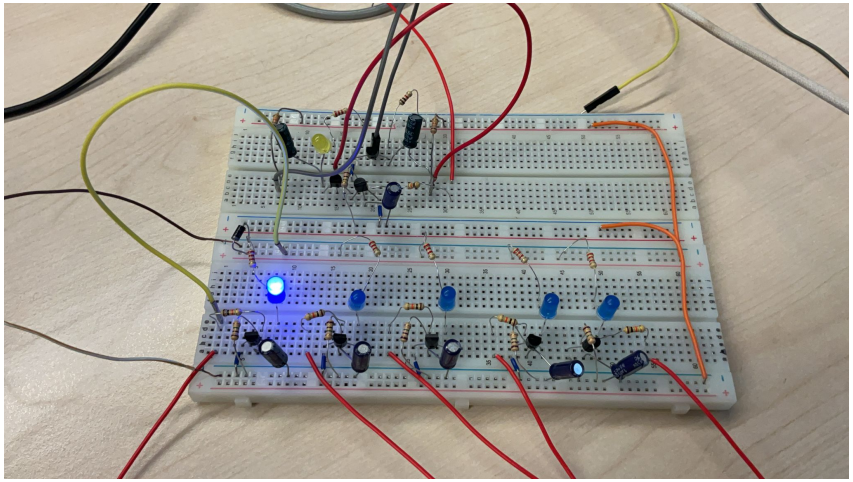
The first five LED currents turn on sequentially and remain close to the intended few-milliampere current range.

$$I_{D_1}, \ldots, I_{D_5} \approx \text{steady low-current indication}$$

Physical meaning

D1 to D5 behave as ordinary steady water-level indicators rather than blinking or oscillating outputs.

Physical Result: LED Current and Activation



Measured LED current

For a representative LED branch, the physical current was in the intended few-milliampere range.

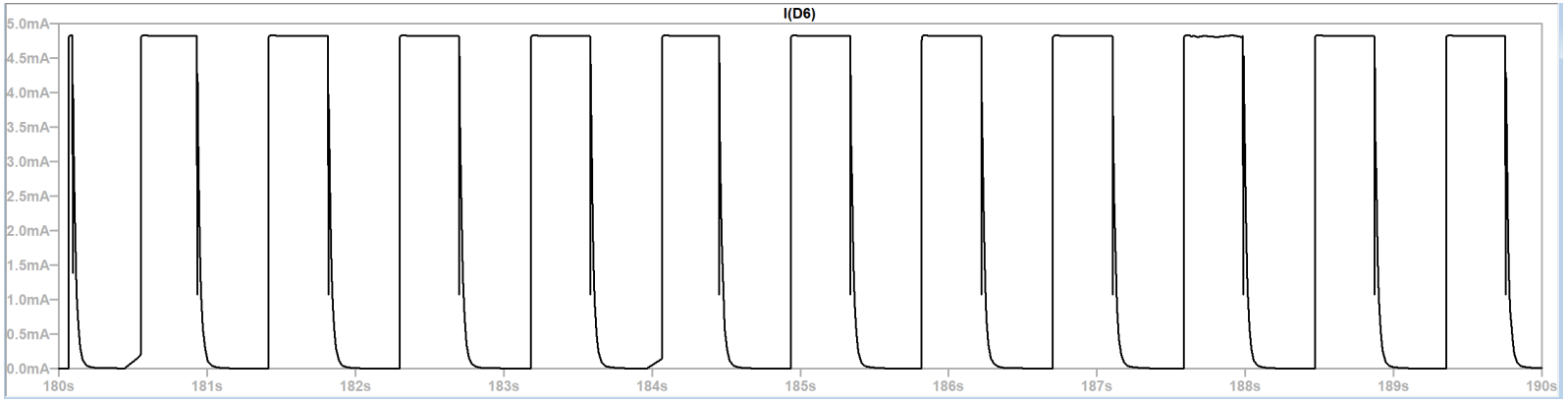
$$I_{\mathrm{LED,calc}} \approx 4.09\,\mathrm{mA}$$

$$I_{\mathrm{LED,meter}} \approx 3.91\,\mathrm{mA}$$

Experimental meaning

The LEDs operate near the designed low-current range, so the indicator is visible without drawing large current.

Simulation 5: Alarm LED Blinking



What the simulation shows

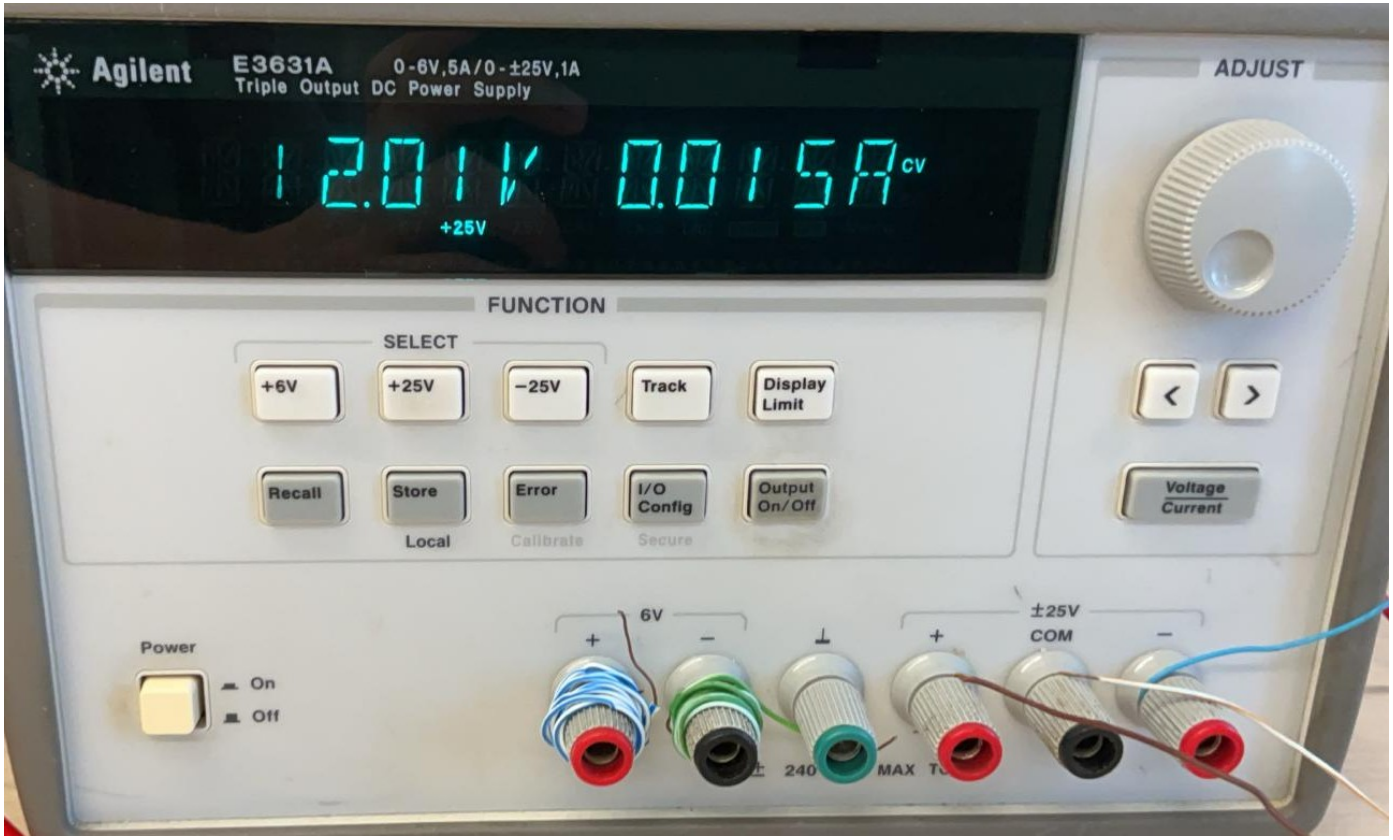
After the alarm-level input is activated, the alarm LED current switches periodically between nearly zero and its ON-state current.

$$P_A \text{ activated} \rightarrow \text{astable enabled} \rightarrow D_A \text{ alternates ON/OFF}$$

Physical meaning

The highest level is not displayed as another steady LED; it is converted into a blinking warning signal.

Physical Result: Alarm Blinking

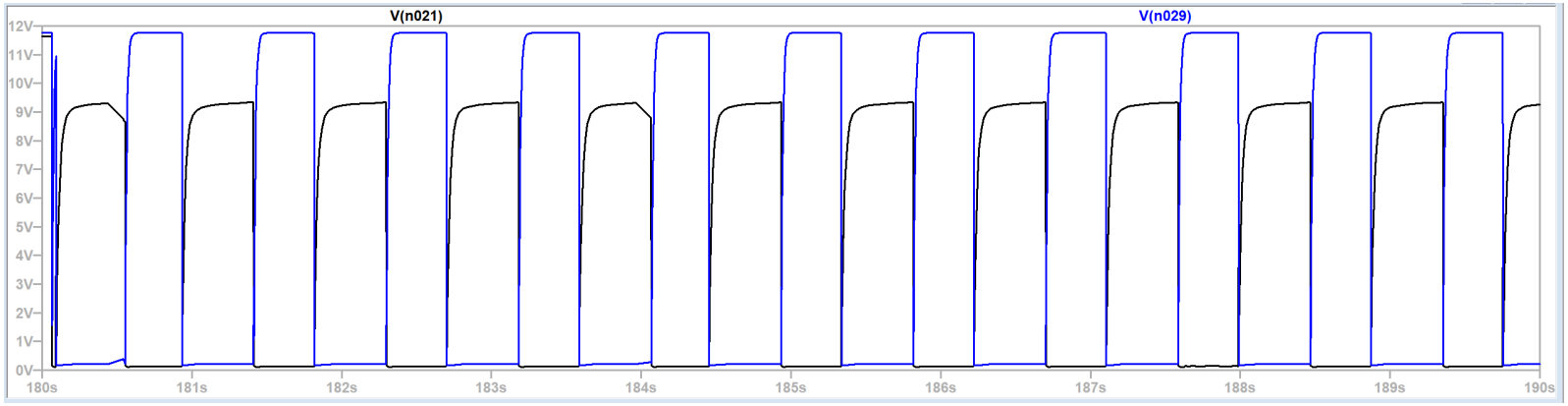


Video verification

The alarm behavior was also verified by video. The video file can be opened from:

`figures/11.17.mp4`

Simulation 6: Astable Multivibrator Operation



What the simulation shows

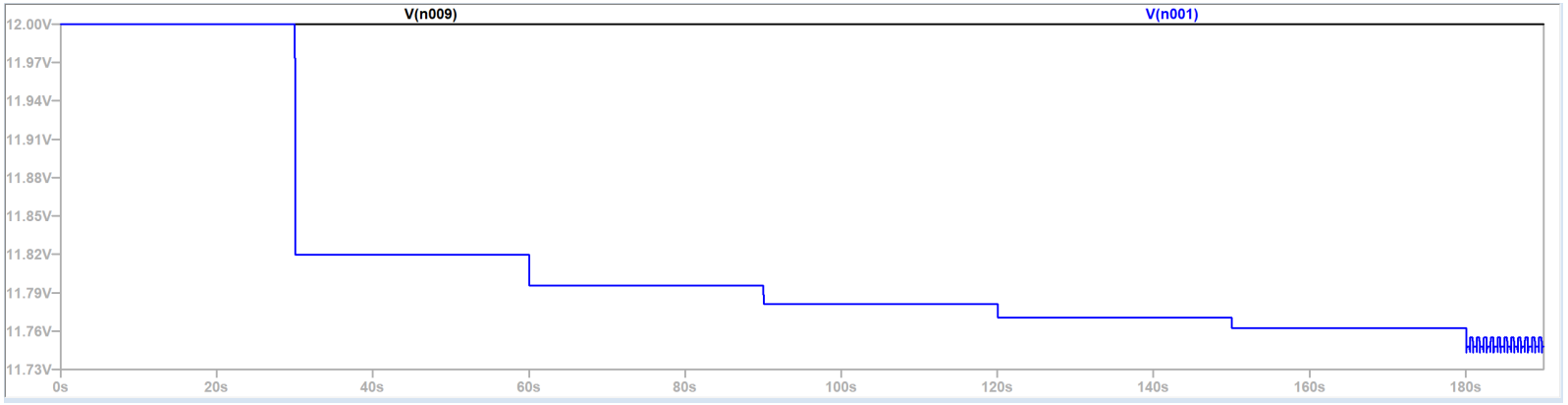
The two astable node voltages alternate periodically between high and low states.

$$Q_{\mathrm{AST1}} \text{ ON} \leftrightarrow Q_{\mathrm{AST2}} \text{ ON}$$

Physical meaning

The blinking alarm is not caused by random flickering. It is produced by the internal oscillation of the astable multivibrator.

Simulation 7: Protected Supply Voltage



What the simulation shows

The protected supply rail remains close to the input supply, with only a small reduction due to the Schottky protection diode and loading.

$$V_{CC} \approx V_{\text{in}} - V_D$$

Physical meaning

The protection diode protects the circuit without removing too much usable voltage from the LED branches and BJT stages.

Physical Result: Protected Rail Measurement

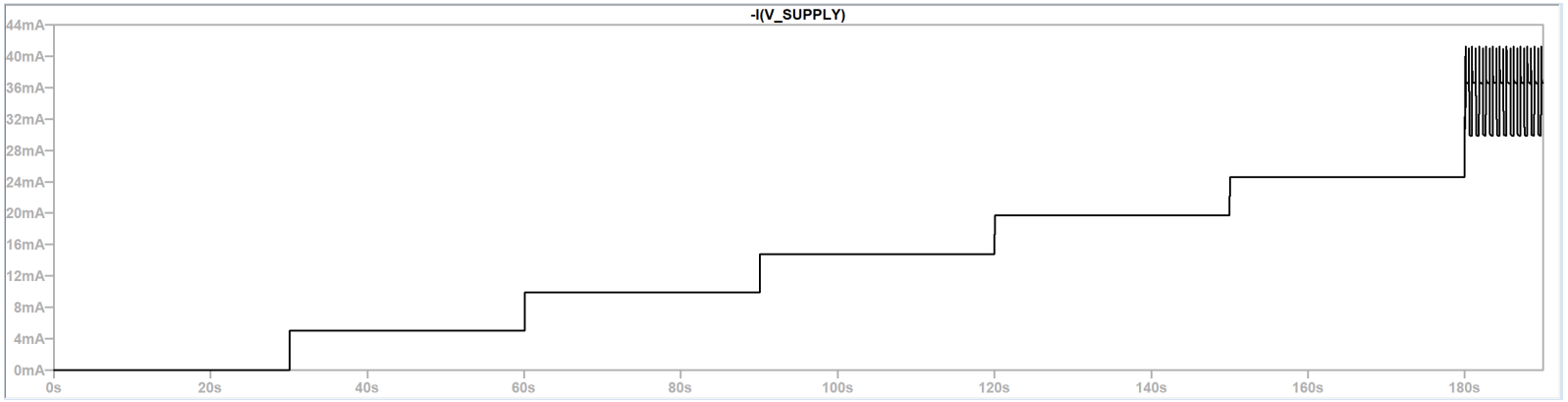


Measured supply values

The physical measurement gave approximately

$$V_{\text{in}} = 12.01\,\text{V}, \qquad V_{CC} = 11.7562\,\text{V}.$$

Simulation 8: Supply Current and Power



What the simulation shows

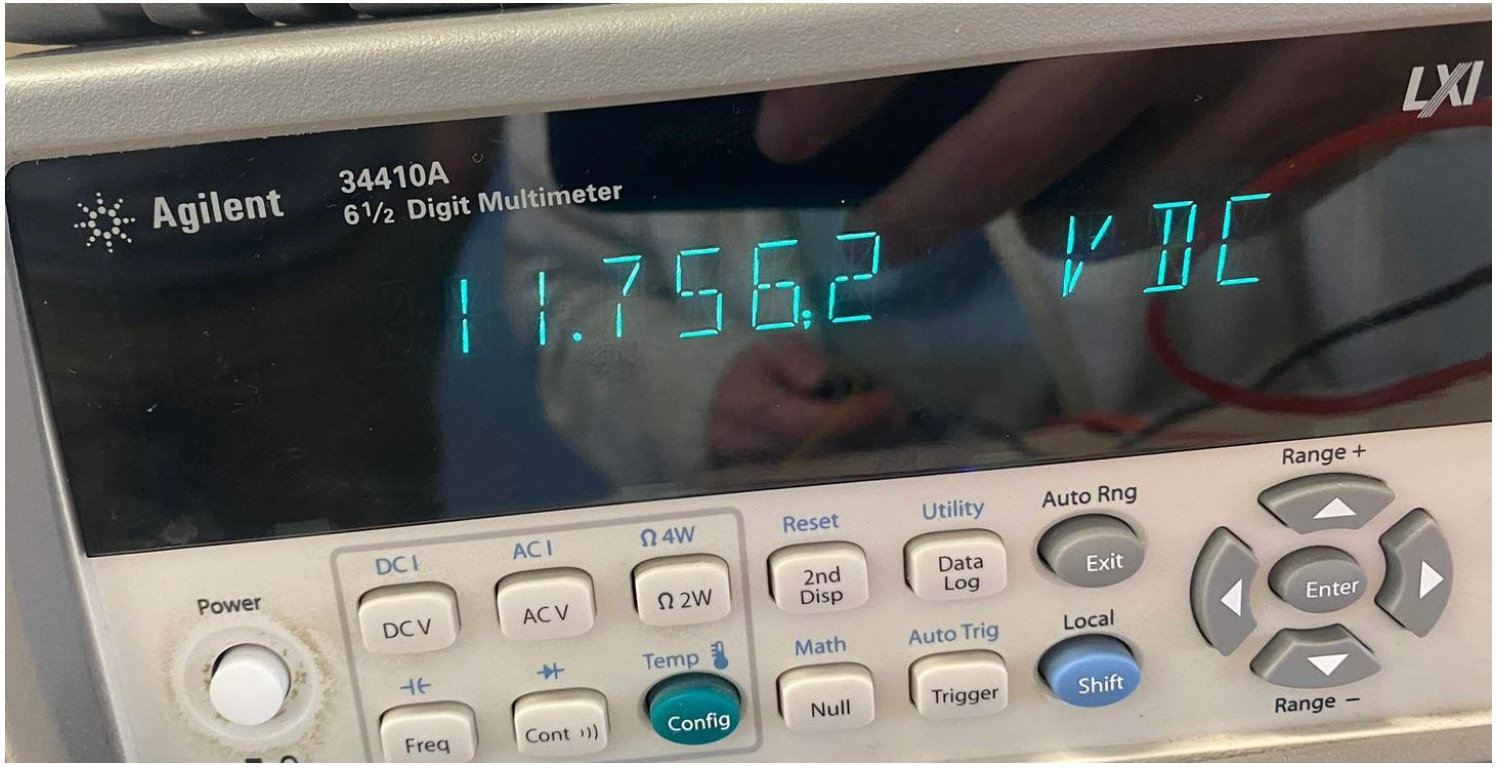
The supply current increases step by step as additional LED branches are activated. After P_A, the astable stage adds periodic current variation.

$$P_{\text{simulated}} < 0.5\,\text{W}$$

Physical meaning

The circuit operates in a low-current and low-power range. Heating is not expected to be the dominant limitation.

Physical Result: Supply Current and Power



Measured trend

The measured supply current increased as more stages became active.

$$I_{P_1} \approx 0.005\,\mathrm{A}, \qquad I_{P_A} \approx 0.038\,\mathrm{A}$$

Simulation and Physical Experiment Comparison

LTspice simulation

- Controlled switch activation
- Fixed modeled water resistance
- Clean waveforms
- Idealized contact behavior

Physical experiment

- Real water conductivity
- Probe contact variation
- Breadboard contact effects
- Component tolerances

water contact $\rightarrow$ BJT switching $\rightarrow$ steady LEDs $\rightarrow$ P_A alarm $\rightarrow$ D_A blinking

Main comparison

Exact numerical agreement is not the main criterion. The important result is that both simulation and physical testing verify the same operating sequence.

Practical Limitations of the Implemented Design

Main limitation

The main limitation is not thermal power dissipation. The main limitation is the sensing method itself: DC conductive sensing through water and metal probes.

- Water conductivity changes with water type, dissolved ions, and impurities.
- Probe-water contact can be unstable due to surface motion or partial wetting.
- Breadboard construction can introduce contact resistance and loose connections.
- Component tolerances affect LED current, RC delay, and blinking frequency.
- DC probe current can cause electrolysis and probe corrosion during long-term use.

Design consequence

MOSFET-Based Improvement Direction I

Why a MOSFET version is considered

The implemented BJT version requires base current through the water path. If water resistance increases, the available base current decreases and switching becomes weaker.

BJT sensing

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_{\text{water}} + R_{\text{BASE}}}$$

$$R_{\text{water}} \uparrow \Rightarrow I_B \downarrow$$

- Current-controlled
- Needs water-path current
- More dependent on conductivity

MOSFET sensing

$$I_G \approx 0$$

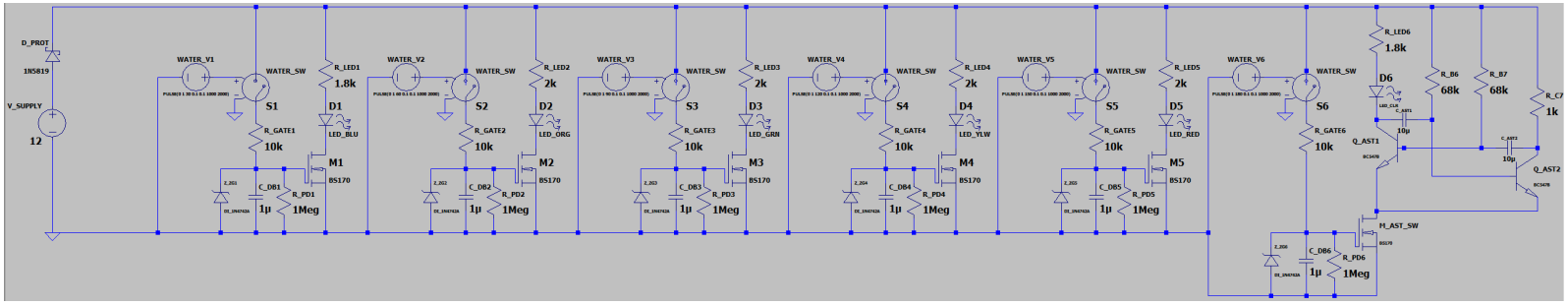
$$V_{GS} > V_{\text{th}}$$

- Voltage-controlled
- Much smaller steady input current
- Less loading of the water path

Improvement idea

The MOSFET version keeps the same output function but improves the sensing interface.

MOSFET-Based Improvement Direction II



Expected advantage

A MOSFET-based sensing stage can reduce continuous sensing current through the water path. This helps reduce electrolysis and probe corrosion risk.

$$V_{DS(\text{on})} \approx I_D R_{DS(\text{on})}$$

Protection requirement

The MOSFET gate must be protected. A gate-source zener clamp can limit excessive $|V_{GS}|$ and protect the gate oxide.

AC-Based Sensing as Future Work

Remaining problem

Even the MOSFET version is still based on conductive-probe sensing. If the sensing signal has a DC component, probe corrosion can still remain a long-term concern.

$$\text{DC probe current} \rightarrow \text{electrolysis} \rightarrow \text{probe corrosion}$$

AC-based idea

A future version could excite the probes using an AC signal instead of continuous DC sensing. This reduces net DC current through the water.

$$\text{AC excitation} \rightarrow \text{reduced net DC current} \rightarrow \text{better long-term durability}$$

Future direction

Final Conclusion

Project result

A six-level conductive-probe-based discrete analog water level indicator was designed, simulated, and physically tested.

- The requirement was at least three detected water levels.
- The implemented design uses five steady indicators and one alarm level.
- LTspice verifies the internal switching, LED currents, alarm blinking, and supply behavior.
- Physical testing confirms the main functional sequence.
- Long-term use would require better probe design and preferably AC-based sensing.

$P_n \rightarrow D_n$ steady indication, $P_A \rightarrow D_A$ blinking alarm

Main conclusion